

Improve and Challenge

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Test Diverse Users

Without opening the old lesson, explain one key idea from Test Diverse Users. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Evidence and Sources

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Test neighbors for Improving without hiding failures. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Repair independently

Write the eligibility predicate for at most 12 minutes and a regression set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

6. Solve the given case

A repair raises success from 7/10 to 9/10 on the same validation set.

What belongs in the improvement claim?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 35 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Publish the cause, one change, fixed test set, before/after evidence, and remaining limitation.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Locate the cause

Why does $\text{minutes} < \text{limit}$ reject a valid exactly-at-limit activity?

My reasoning and evidence

13. W2 Test neighbors

For limit 15, give expected eligibility at 14, 15 and 16.

My reasoning and evidence

14. W3 Reject a bad fix

A proposed fix accepts $\text{minutes} \leq \text{limit} + 1$. Give a counterexample at limit 15.

My reasoning and evidence

15. W4 Repair independently

Write the eligibility predicate for at most 12 minutes and a regression set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

My reasoning and evidence

16. W5 Write the change log

Record the bug, change, evidence and one remaining limit without claiming all bugs are gone.

My reasoning and evidence

17. Testing functions and interfaces — W1 Choose test scale

Classify testing score(3,2) versus running raw rows through the whole chooser.

My reasoning and evidence

18. Testing functions and interfaces — W2 Calculate an oracle

What expected score should a test use for fun 4 and learning 3?

My reasoning and evidence

19. Testing functions and interfaces — W3 Repair a weak test

A test compares `recommend(items,10)` to itself. Replace it with a useful assertion.

My reasoning and evidence

20. Testing functions and interfaces — W4 Test independently

Design an integration fixture: A valid (minutes 10, fun 2, learning 4), B valid (10,3,2), C invalid (5,9,1), limit 10. State winner, counts and a reversed-order expectation.

My reasoning and evidence

21. Testing functions and interfaces — W5 Cover absence

What expected result should an empty-input integration case specify?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Improve and Challenge

A fictional recommender excludes books at an intended including equality page limit.

Contract: pages $\leq$ 100; highest interest; ties choose smaller ID. Old code uses pages $<$ 100.

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

Improve and Challenge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify cause

A wrongly excluded at equality, so old code selects B.



2. Focused repair

Replace <100 with ≤ 100 without changing ranking or type checks.



3. Expected revised result

A becomes eligible and wins; C remains excluded.



4. Challenge repair

Rerun 99,100,101; invalid pages; tied scores; reversed row order.

CHANGE ONE THING

An alternative patch simply increases the limit to 101.

Explain the changed result using the matching lesson.

Why are tests at both 100 and 101 needed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For each recommender revision, retain cause, code change, targeted test, regression results and unresolved issues.

24. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For each recommender revision, retain cause, code change, targeted test, regression results and unresolved issues.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender excludes books at an intended including equality page limit.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Rerun 99,100,101; invalid pages; tied scores; reversed row order.

1. Identify cause
2. Expected revised result
3. Challenge repair
4. Focused repair

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Challenge repair Rerun 99,100,101; invalid pages; tied scores; reversed row order. A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. Regression tests check surrounding behavior that should remain unchanged. A tie case and reversed input order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Practice 2: explain the deciding evidence

Changed-case practice: An alternative patch simply increases the limit to 101. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. C is wrongly admitted as well as A.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: C is wrongly admitted as well as A. Changing the maximum alters the contract instead of repairing its comparison.

Repair a missing building block

Review Test Diverse Users (ai-2/34). Goal: Evaluate across varied cases. Short practice: Can these records establish performance for different user groups?. Return to this week and explain what you changed.

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review Evidence and Sources (ai-2/29). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
